THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 4: **Journey Inside the Atom** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An atom of sodium has atomic number 11 and mass number 23. The number of neutrons in it is: 1

(a) 23 **(b)** 13 **(c)** 11 **(d)** 12

2. An atom of iron has atomic number 26 and mass number 56. The number of neutrons in it is: 1

(a) 30 **(b)** 31 **(c)** 56 **(d)** 26

3. The electronic configuration (K, L, M, N shells) of the sulfur atom (Z = 16) is: 1

(a) 2, 8, 6 **(b)** 8, 2, 6 **(c)** 2, 9, 5 **(d)** 2, 8, 7

4. The valency of magnesium (Z = 12) is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 3

5. The maximum number of electrons that the shell with n = 3 can hold is: 1

(a) 20 **(b)** 36 **(c)** 9 **(d)** 18

6. The Greek philosophers who proposed indivisible particles called ‘atomos’ were: 1

(a) Thomson and Rutherford **(b)** Leucippus and Democritus
(c) Dalton and Proust **(d)** Bohr and Bury

7. A few α-particles rebounded at large angles. This led Rutherford to conclude that: 1

(a) electrons have no charge
(b) neutrons exist
(c) positive charge is spread evenly
(d) the positive charge is concentrated in a tiny dense nucleus

8. The neutron was discovered by: 1

(a) Niels Bohr **(b)** J. J. Thomson **(c)** James Chadwick **(d)** Goldstein

9. The mass number of an atom is the sum of its: 1

(a) electrons and neutrons **(b)** protons and electrons
(c) protons and neutrons **(d)** shells and electrons

10. The isotope used for determining the age of fossils is: 1

(a) carbon-14 **(b)** cobalt-60 **(c)** iodine-131 **(d)** uranium-235

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Name the ancient Indian and Greek thinkers who proposed that matter is made of indivisible particles. 2

12. Why did Thomson conclude that electrons are present in all atoms? 2

13. Describe Thomson's plum pudding model. 2

14. What did most α-particles do in the gold foil experiment and what does it show? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain the concept of the atom in ancient India and Greece and how Dalton's idea differed from it. 3

16. Describe Thomson's cathode ray experiment and what he concluded. 3

17. Why was Thomson's model rejected after the gold foil experiment? 3

OR

Describe the gold foil experiment and its observations. 3

18. State the main features of Rutherford's nuclear (planetary) model. 3

OR

If the diameter of an atom is 10^{-10} m and of its nucleus 10^{-15} m, how many times smaller is the nucleus? Use the cricket ground analogy. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe Thomson's cathode ray tube experiment. 5
(ii) What did he conclude?
(iii) Describe the plum pudding model and why it was proposed.

OR

(i) Describe the gold foil experiment with observations. 5
(ii) State the conclusions drawn.
(iii) What would you expect if a thicker foil were used?

20. **Case study:** If an atom were the size of a cricket ground (100 m), its nucleus would be a black pepper grain. 5
(i) How many times smaller is the nucleus than the atom?
(ii) Which part contains nearly all the mass?
(iii) What does this tell us about the structure of the atom?
(iv) Why did Rutherford's model fail to explain stability?

OR

(i) Describe Bohr's model. 5
(ii) How does it explain the stability of atoms?
(iii) Why are the shells called K, L, M and N?

END OF QUESTION PAPER — Total: 40 Marks